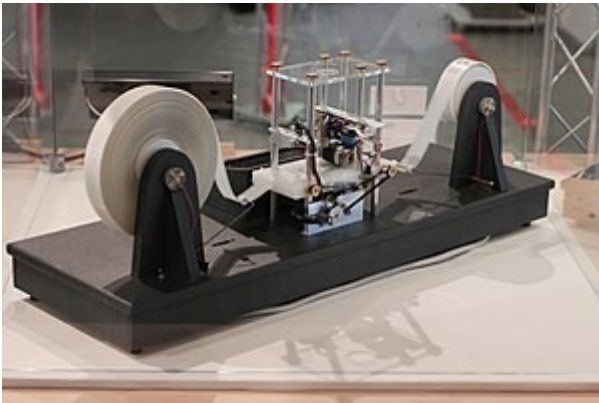


Church-Turing Thesis

"a function on the natural numbers can be calculated by an effective method if and only if it is computable by a Turing machine."

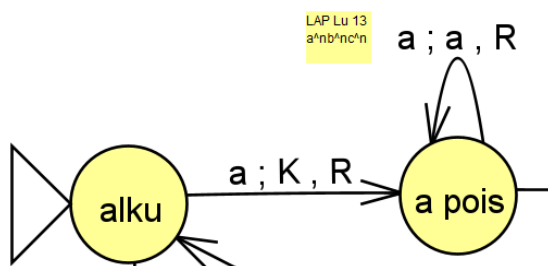


Alonzo Church was Turing's dissertation advisor. Church developed the so-called Lambda calculus, which was a model corresponding to Turing's machine but significantly more complex and difficult compared to Turing's machine.

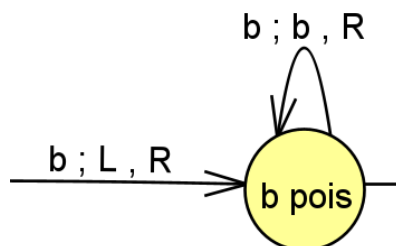
Together they developed the above-mentioned thesis, which is based on Turing's machine.

Lecture 9 dealt with the grammar for the language $\{a^n b^n c^n\}$. The corresponding Turing machine. Note! This differs slightly from what was covered in the lectures, but in terms of operation and principle it is the same.

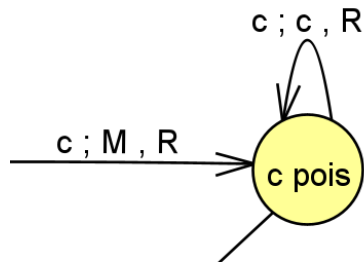
- 1) Remove the first a and skip over the rest



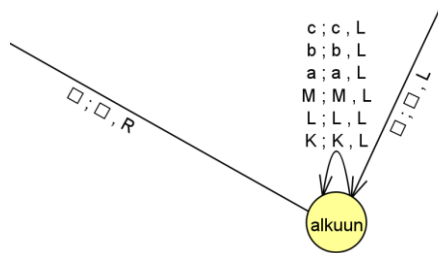
- 2) Remove the first b and skip over the rest



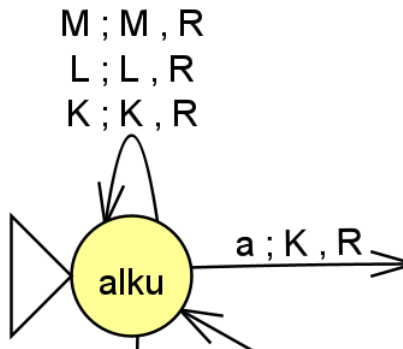
- 3) The same with c



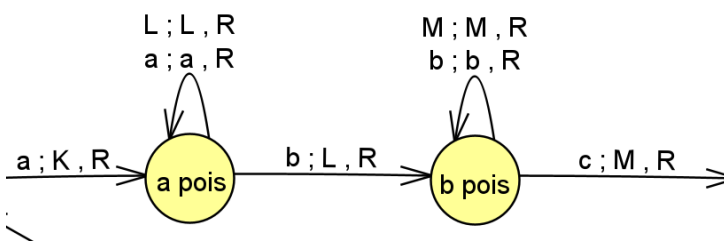
4) Return to the end of the word



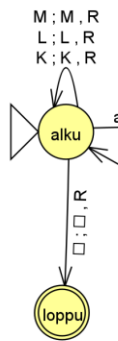
5) Go past the M:s, until we reach an a. If an a is not found, go past Ls and Ms.



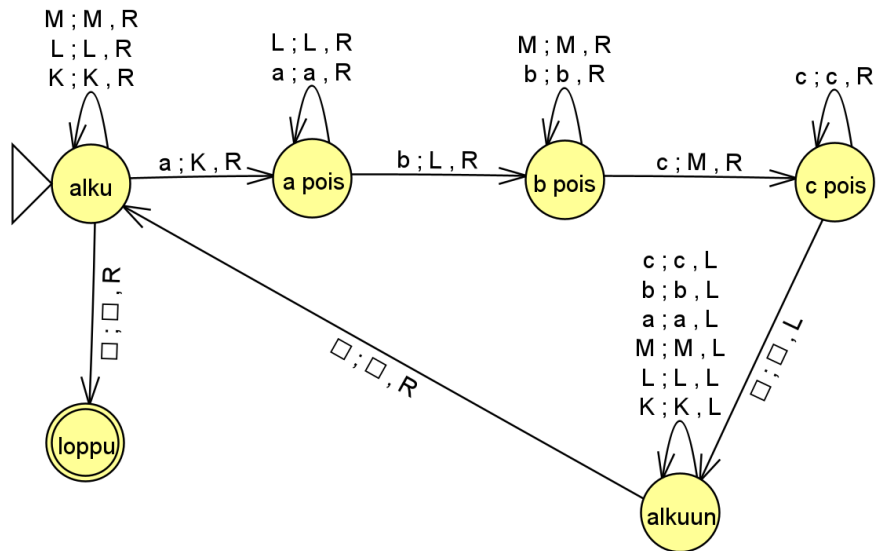
6) Repeat and jump over the marked a (K) and b(L)



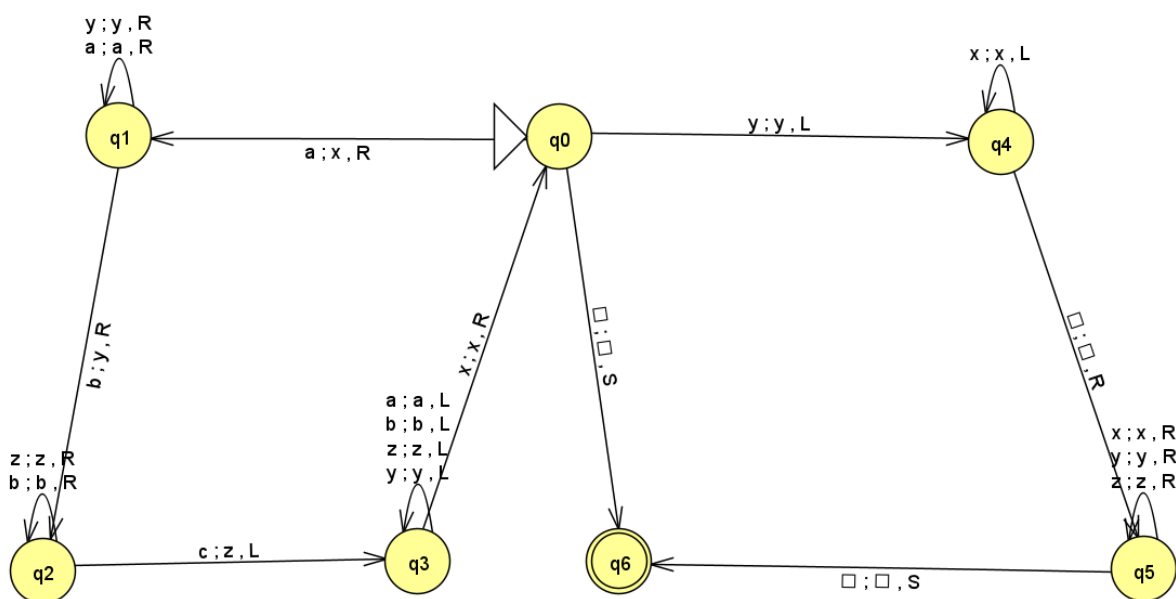
7) If the end of the tape was reached without an a (tape contained only letters K,L and M), enter the final state



Final result:



Version made on lectures(minor differences in the principle, this visits the end and the beginning of the input less often)



Define the grammar for $\{a^n b^n c^n \mid n \in \mathbb{N}\}$ (0 belonging to $\mathbb{N}$)

Possible words include

abc

aabbcc

aaabbbccc

$S \rightarrow abc \mid aAbc$

$Ab \rightarrow bA$

$Ac \rightarrow Bbcc$

$bB \rightarrow Bb$

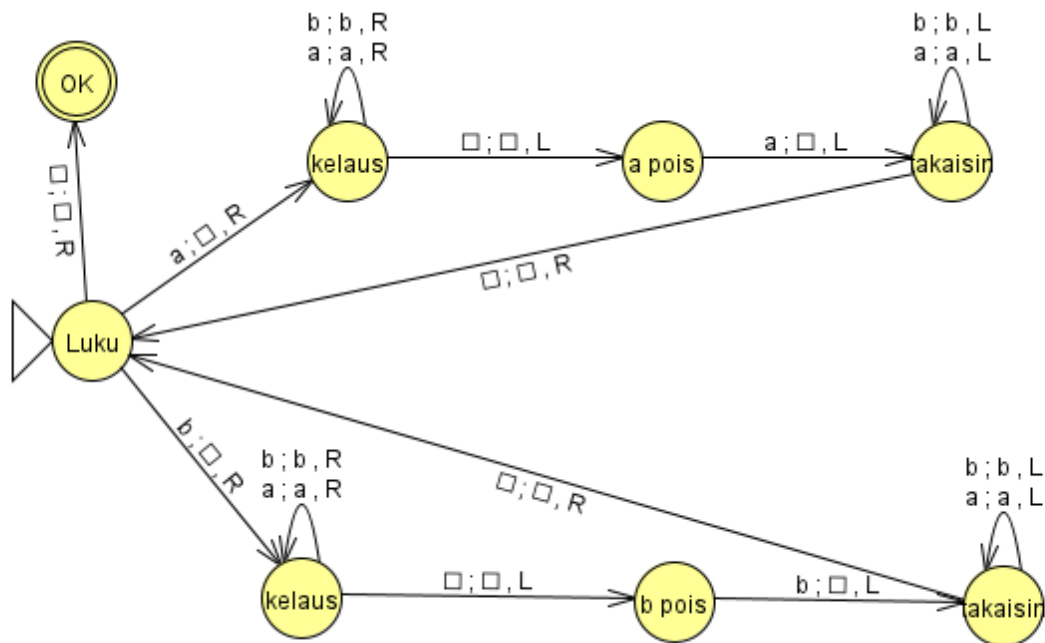
$aB \rightarrow aa \mid aaA$

johdetaan aaabbbccc

$S \rightarrow aAbc \rightarrow abAc \rightarrow abBbcc \rightarrow aBbcc \rightarrow aaAbcc \rightarrow aabAbcc \rightarrow aabbAcc \rightarrow aabbBbcc \rightarrow aabBbbccc \rightarrow aaBbbbccc \rightarrow aaabbbccc$

Palindrome Turing, alphabet a,b

A Turing machine for a palindrome in alphabet a,b e.g. abba



Time requirement?

n charactes

n removals

n word travels

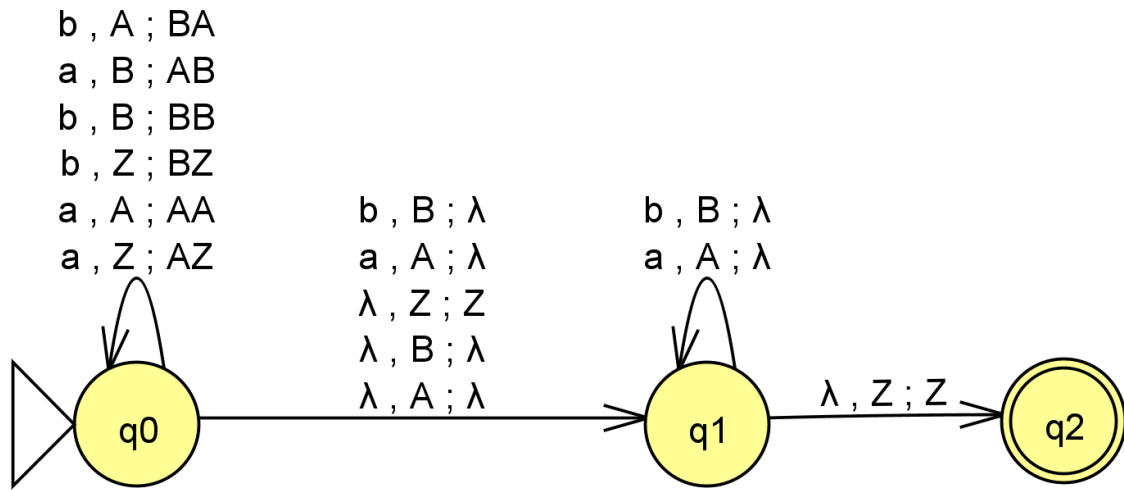
→ $O(n^2)$

The grammar for this palindrome Turing machine?

Simple

$S \rightarrow aSa \mid bSb \mid \epsilon$

Is there any context? No, hence it does not need a Turing machine. It is context-free. Is it regular? No, because for each a You need a b. Hence, we should be able to do it with a PDA. The following is nondeterministic PDA for the task (the Turing machine is of course deterministic).



Context-free grammar $a^k b^l c^{k+l}$ (PDA)

$S \rightarrow \lambda$

$S \rightarrow aMC$

$M \rightarrow aMC$

$M \rightarrow aLC$

$M \rightarrow bLC$

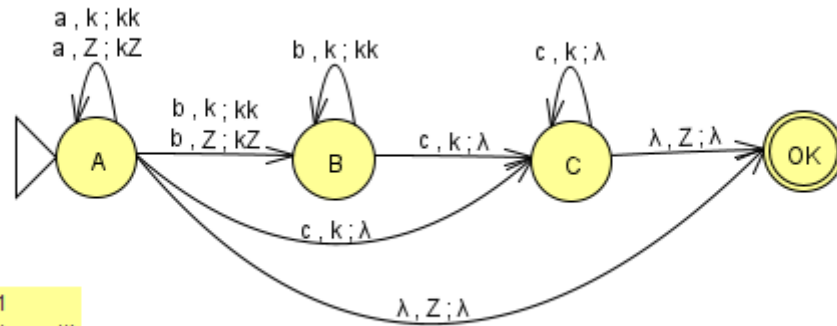
$C \rightarrow c$

$M \rightarrow \lambda$

$L \rightarrow \lambda$

$S \rightarrow bLC$

$L \rightarrow bLC$



HA6 T1
Pinoautomaatti
 $a^k b^l c^{k+l}$

aabbccccc

$S \rightarrow a a b b c c c c c$